

# Probability and Statistics

## Distributions of Several Random Variables

*Dr. Yasir Ali (yali@ceme.nust.edu.pk)*

DBS&H, CEME-NUST

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Our study of random variables and their probability distributions in the preceding sections is restricted to one-dimensional sample spaces, in that we recorded outcomes of an experiment as values assumed by a single random variable. There will be situations, however, where we may find it desirable to record the simultaneous outcomes of several random variables. For example, we might measure the amount of precipitate  $P$  and volume  $V$  of gas released from a controlled chemical experiment, giving rise to a two-dimensional sample space consisting of the outcomes  $(p, v)$ , or we might be interested in the hardness  $H$  and tensile strength  $T$  of cold-drawn copper, resulting in the outcomes  $(h, t)$ . In a study to determine the likelihood of success in college based on high school data, we might use a three dimensional sample space and record for each individual his or her aptitude test score, high school class rank, and grade-point average at the end of freshman year in college.

# Joint Probability Distribution

The function  $f(x, y)$  is a joint probability distribution or probability mass function of the discrete random variables  $X$  and  $Y$  if

- ①  $f(x, y) \geq 0$  for all  $(x, y)$ ,
- ②  $\sum_x \sum_y f(x, y) = 1$ ,
- ③  $P(X = x, Y = y) = f(x, y)$ .

For any region  $A$  in the  $xy$  plane,  $P[(X, Y) \in A] = \sum \sum_A f(x, y)$ .

# Joint Density Function

The function  $f(x, y)$  is a joint density function of the continuous random variables  $X$  and  $Y$  if

- 1  $f(x, y) \geq 0$  for all  $(x, y)$ ,
- 2  $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$ ,
- 3  $P[(X, Y) \in A] = \int \int_A f(x, y) dx dy$ , for any region  $A$  in the  $xy$  plane

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Let  $S$  be the sample space when two fair dice are tossed, and let  $X$  and  $Y$  be the random variables on  $S$ . That is to each  $(a, b) \in S$ ,  $X$  assigns the maximum of numbers,  $Y$  assigns the sum of the numbers:

$$X(a, b) = \max(a, b) \quad \text{and} \quad Y(a, b) = a + b$$

|   | 2              | 3              | 4              | 5              | 6              | 7              | 8              | 9              | 10             | 11             | 12             | sum             |
|---|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|-----------------|
| 1 | $\frac{1}{36}$ | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | $\frac{1}{36}$  |
| 2 | 0              | $\frac{2}{36}$ | $\frac{1}{36}$ | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | $\frac{3}{36}$  |
| 3 | 0              | 0              | $\frac{2}{36}$ | $\frac{2}{36}$ | $\frac{1}{36}$ | 0              | 0              | 0              | 0              | 0              | 0              | $\frac{5}{36}$  |
| 4 | 0              | 0              | 0              | $\frac{2}{36}$ | $\frac{2}{36}$ | $\frac{2}{36}$ | $\frac{1}{36}$ | 0              | 0              | 0              | 0              | $\frac{7}{36}$  |
| 5 | 0              | 0              | 0              | 0              | $\frac{2}{36}$ | $\frac{2}{36}$ | $\frac{2}{36}$ | $\frac{2}{36}$ | $\frac{1}{36}$ | 0              | 0              | $\frac{9}{36}$  |
| 6 | 0              | 0              | 0              | 0              | 0              | $\frac{2}{36}$ | $\frac{2}{36}$ | $\frac{2}{36}$ | $\frac{2}{36}$ | $\frac{2}{36}$ | $\frac{1}{36}$ | $\frac{11}{36}$ |
|   | $\frac{1}{36}$ | $\frac{2}{36}$ | $\frac{3}{36}$ | $\frac{4}{36}$ | $\frac{5}{36}$ | $\frac{6}{36}$ | $\frac{5}{36}$ | $\frac{4}{36}$ | $\frac{3}{36}$ | $\frac{2}{36}$ | $\frac{1}{36}$ |                 |

Two ballpoint pens are selected at random from a box that contains 3 blue pens, 2 red pens, and 3 green pens. If  $X$  is the number of blue pens selected and  $Y$  is the number of red pens selected, find

- (a) the joint probability function  $f(x, y)$ ,
- (b)  $P[(X, Y) \in A]$ , where  $A$  is the region  $\{(x, y) | x + y \leq 1\}$ .

The possible pairs of values  $(x, y)$  are  $(0, 0)$ ,  $(0, 1)$ ,  $(1, 0)$ ,  $(1, 1)$ ,  $(0, 2)$ , and  $(2, 0)$ .

$(0,1)$  represents

- 1 No blue pen
- 2 One red pen
- 3 One green pen

The total number of equally likely ways of selecting any 2 pens from the 8 is  $C_2^8 = 28$ .

The number of ways of selecting 1 red from 2 red pens and 1 green from 3 green pens is

$$C_1^2 \cdot C_1^3 = 6 \quad \text{Hence, } f(0, 1) = 6/28 = 3/14.$$



(a) Joint probability distribution can be represented by the formula

$$f(x, y) = \frac{{}^3C_x {}^2C_y {}^3C_{2-x-y}}{{}_2C_8}, \text{ for } x = 0, 1, 2, y = 0, 1, 2 \text{ and } 0 \leq x + y \leq 2$$

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$$f(x, y) = \frac{{}^3C_x {}^2C_y {}^3C_{2-x-y}}{{}^8C_2}, \text{ for } x = 0, 1, 2, y = 0, 1, 2 \text{ and } 0 \leq x + y \leq 2$$

| $f(x, y)$ |   | $x$            |                 |                | $h(y)$          |
|-----------|---|----------------|-----------------|----------------|-----------------|
|           |   | 0              | 1               | 2              |                 |
| $y$       | 0 | $\frac{3}{28}$ | $\frac{9}{28}$  | $\frac{3}{28}$ | $\frac{15}{28}$ |
|           | 1 | $\frac{3}{14}$ | $\frac{3}{14}$  | 0              | $\frac{3}{7}$   |
|           | 2 | $\frac{1}{28}$ | 0               | 0              | $\frac{1}{28}$  |
| $g(x)$    |   | $\frac{5}{14}$ | $\frac{15}{28}$ | $\frac{3}{28}$ | 1               |

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(b)  $P(X + Y \leq 1) = f(0, 0) + f(0, 1) + f(1, 0) = \frac{9}{14}$

## Example

A privately owned business operates both a drive-in facility and a walk-in facility. On a randomly selected day, let  $X$  and  $Y$ , respectively, **be the proportions of the time that the drive-in and the walk-in facilities are in use**, and suppose that the joint density function of these random variables is

$$f(x, y) = \begin{cases} \frac{2}{5} (2x + 3y), & 0 \leq x \leq 1, 0 \leq y \leq 1; \\ 0, & \text{otherwise.} \end{cases}$$

(a) Verify condition 2.

(b) find  $P[(X, Y) \in A]$ , where  $A = \{(x, y) | 0 \leq x \leq \frac{1}{2}, \frac{1}{4} \leq y \leq \frac{1}{2}\}$

$$(a) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = \int_0^1 \int_0^1 \frac{2}{5} (2x + 3y) = 1$$

$$(b) P[(X, Y) \in A] = \int_0^{\frac{1}{2}} \int_{\frac{1}{4}}^{\frac{1}{2}} \frac{2}{5} (2x + 3y) dx dy$$

# Marginal Distributions

The marginal distributions of  $X$  alone and of  $Y$  alone are

$$g(x) = \sum_y f(x, y) \quad \text{and} \quad h(y) = \sum_x f(x, y)$$

for the discrete case and

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy \quad \text{and} \quad h(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

for the continuous case.

Given the joint probability distribution  $f(x, y)$  of the discrete random variables  $X$  and  $Y$ , the probability distribution  $g(x)$  of  $X$  alone is obtained by summing  $f(x, y)$  over the values of  $Y$ .

We define  $g(x)$  and  $h(y)$  to be the marginal distributions of  $X$  and  $Y$ , respectively. When  $X$  and  $Y$  are continuous random variables, summations are replaced by integrals.

### Example

Find  $g(x)$  and  $h(y)$  for the joint density function

$$f(x, y) = \begin{cases} \frac{2}{5} (2x + 3y), & 0 \leq x \leq 1, 0 \leq y \leq 1; \\ 0, & \text{otherwise.} \end{cases}$$

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^1 \frac{2}{5} (2x + 3y) dy = \frac{4x + 3}{5}$$

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^1 \frac{2}{5} (2x + 3y) dx = \frac{2(1 + 3y)}{5}$$

# Conditional Probability

$$P(A \cap B) = P(A) \cdot P(B/A)$$

Let  $X$  and  $Y$  be two random variables, discrete or continuous. The conditional distribution of the random variable  $Y$  given that  $X = x$  is

$$f(y|x) = \frac{f(x, y)}{g(x)}, \text{ provided } g(x) > 0$$

Similarly, the conditional distribution of  $X$  given that  $Y = y$  is

$$f(x|y) = \frac{f(x, y)}{h(y)}, \text{ provided } h(y) > 0$$

## Example

Two ballpoint pens are selected at random from a box that contains 3 blue pens, 2 red pens, and 3 green pens. If  $X$  is the number of blue pens selected and  $Y$  is the number of red pens selected, find the conditional distribution of  $X$ , given that  $Y = 1$ , and use it to determine  $P(X = 0 \mid Y = 1)$ .

$$f(x, y) = \frac{{}^3C_x {}^2C_y {}^3C_{2-x-y}}{{}^8C_2}, \text{ for } x = 0, 1, 2, y = 0, 1, 2 \text{ and } 0 \leq x + y \leq 2$$

| $f(x, y)$ |   | $x$            |                 |                | $h(y)$          |
|-----------|---|----------------|-----------------|----------------|-----------------|
|           |   | 0              | 1               | 2              |                 |
| $y$       | 0 | $\frac{3}{28}$ | $\frac{9}{28}$  | $\frac{3}{28}$ | $\frac{15}{28}$ |
|           | 1 | $\frac{3}{14}$ | $\frac{3}{14}$  | 0              | $\frac{3}{7}$   |
|           | 2 | $\frac{1}{28}$ | 0               | 0              | $\frac{1}{28}$  |
| $g(x)$    |   | $\frac{5}{14}$ | $\frac{15}{28}$ | $\frac{3}{28}$ | 1               |



|        |   | $x$            |                 |                | $h(y)$          |
|--------|---|----------------|-----------------|----------------|-----------------|
|        |   | 0              | 1               | 2              |                 |
| $y$    | 0 | $\frac{3}{28}$ | $\frac{9}{28}$  | $\frac{3}{28}$ | $\frac{15}{28}$ |
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First we need to find  $g(x)$  and  $h(y)$ .

| $x$    | 0              | 1               | 2              |
|--------|----------------|-----------------|----------------|
| $g(x)$ | $\frac{5}{14}$ | $\frac{15}{28}$ | $\frac{3}{28}$ |

| $y$    | 0               | 1             | 2              |
|--------|-----------------|---------------|----------------|
| $h(y)$ | $\frac{15}{28}$ | $\frac{3}{7}$ | $\frac{1}{28}$ |

$$f(x|1) = \frac{f(x, 1)}{h(1)}, \text{ here } h(1) = \frac{3}{7} \implies f(x|1) = (7/3) f(x, 1)$$

$$f(0|1) = (7/3) f(0, 1) = (7/3) \frac{{}^3C_0 {}^2C_1}{{}_2C_2} = (7/3) (3/14) = \frac{1}{2}$$

$$\mu = \sum_x xf(x) \quad (\text{Discrete distribution})$$

$$\mu = \int_{-\infty}^{\infty} xf(x)dx \quad (\text{Continuous distribution})$$

$$E[g(\textcolor{red}{X}, Y)] = \begin{cases} \sum_{\textcolor{red}{x}} \sum_{\textcolor{blue}{y}} g(x, y) f(x, y) & \text{Discrete Case} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) \textcolor{red}{d}x \textcolor{blue}{d}y & \text{Continuous Case} \end{cases}$$

$$\sigma^2 = E[(X - \mu)^2] = \sum_x (x - \mu)^2 f(x) \quad (\text{Discrete distribution})$$

$$= \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx \quad (\text{Continuous distribution})$$

$$\sigma_{g(X)}^2 = E[(g(X) - \mu_{g(X)})^2] = \sum_x (g(X) - \mu_{g(X)})^2 f(x) \quad (\text{Dis.})$$

$$= \int_{-\infty}^{\infty} (g(X) - \mu_{g(X)})^2 f(x) dx \quad (\text{Cont.})$$

Here

$$\mu_{g(X)} = E[g(X)]$$

# Covariance of X and Y

$$\sigma_{\textcolor{red}{X}\textcolor{blue}{X}}^2 = E[(X - \mu)^2] = \sum_x (x - \mu)^2 f(x) \quad \text{(Discrete distribution)}$$

$$= \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx \quad \text{(Continuous distribution)}$$

$$\boxed{\sigma_{\textcolor{red}{X}\textcolor{blue}{Y}}} = ?$$

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$$= \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx \quad \text{(Continuous distribution)}$$

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# Covariance of X and Y

Let X and Y be random variables with joint probability distribution  $f(x,y)$ .  
The **covariance of X and Y** is

$$\begin{aligned}\sigma_{XY} &= E[(X - \mu_X)(Y - \mu_Y)] \\ &= \sum_x \sum_y (X - \mu_X)(Y - \mu_Y) f(x, y) \quad (\text{Dis.}) \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (X - \mu_X)(Y - \mu_Y) f(x, y) dx dy \quad (\text{Cont.})\end{aligned}$$

Here

$$\mu_X = E[X] \quad \text{and} \quad \mu_Y = E[Y]$$

$$\sigma_{XY} = E(XY) - \mu_X \mu_Y$$

$$\sigma_{XY} = E(XY) - \mu_X \mu_Y$$

By definition  $\sigma_{XY} = \sum_x \sum_y (X - \mu_X)(Y - \mu_Y) f(x, y)$

$$\begin{aligned}\sigma_{XY} &= \sum_x \sum_y (XY - \mu_X Y - \mu_Y X + \mu_X \mu_Y) f(x, y) \\ &= \sum_x \sum_y XY f(x, y) - \mu_X \sum_x \sum_y Y f(x, y) - \mu_Y \sum_x \sum_y X f(x, y) \\ &\quad + \mu_X \mu_Y \sum_x \sum_y f(x, y)\end{aligned}$$

$$\sum_x \sum_y y f(x, y) = \sum_y y g(y) = \mu_Y \text{ similarly, } \sum_x \sum_y x f(x, y) = \mu_X$$

$$\begin{aligned}\sigma_{XY} &= E[XY] - \mu_X \mu_Y - \mu_X \mu_Y + \mu_X \mu_Y \\ &= E[XY] - \mu_X \mu_Y\end{aligned}$$

Two ballpoint pens are selected at random from a box that contains 3 blue pens, 2 red pens, and 3 green pens. If  $X$  is the number of blue pens selected and  $Y$  is the number of red pens selected, find the covariance of  $X$  and  $Y$ .

$$\sigma_{XY} = E[XY] - \mu_X \mu_Y$$

We know that

$$\begin{aligned} E[XY] &= \sum_{x=0}^2 \sum_{y=0}^2 xyf(x, y) = (0)(0)f(0, 0) + (0)(1)f(0, 1) + (1)(0)f(1, 0) \\ &\quad + (1)(1)f(1, 1) + (2)(0)f(2, 0) = \\ \mu_X &= \sum_x xg(x) = 3/4 \text{ and } \mu_Y = \sum_y yh(y) = 1/2 \\ \sigma_{XY} &= E[XY] - \mu_X \mu_Y = \frac{3}{14} - \frac{3}{8} \end{aligned}$$



# Correlation Coefficient

There is a scale-free version of the covariance called the correlation coefficient that is used widely in statistics.

Let  $X$  and  $Y$  be random variables with covariance  $\sigma_{XY}$  and standard deviations  $\sigma_X$  and  $\sigma_Y$ , respectively. The **correlation coefficient** of  $X$  and  $Y$  is

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \cdot \sigma_Y}$$

It should be clear to the reader that  $\rho_{XY}$  is free of the units of  $X$  and  $Y$ . The correlation coefficient satisfies the inequality  $-1 \leq \rho_{XY} \leq 1$ .

Find the correlation coefficient between X and Y in Example

| $f(x, y)$ |   | $x$            |                 |                | $h(y)$          |
|-----------|---|----------------|-----------------|----------------|-----------------|
|           |   | 0              | 1               | 2              |                 |
| $y$       | 0 | $\frac{3}{28}$ | $\frac{9}{28}$  | $\frac{3}{28}$ | $\frac{15}{28}$ |
|           | 1 | $\frac{3}{14}$ | $\frac{3}{14}$  | 0              | $\frac{3}{7}$   |
|           | 2 | $\frac{1}{28}$ | 0               | 0              | $\frac{1}{28}$  |
| $g(x)$    |   | $\frac{5}{14}$ | $\frac{15}{28}$ | $\frac{3}{28}$ | 1               |

We need to calculate  $\sigma_X^2 = E[X^2] - \mu_X^2$  and  $\sigma_Y^2 = E[Y^2] - \mu_Y^2$ .

Find the correlation coefficient between X and Y in Example

|        |   | $x$            |                 |                | $h(y)$          |
|--------|---|----------------|-----------------|----------------|-----------------|
|        |   | 0              | 1               | 2              |                 |
| $y$    | 0 | $\frac{3}{28}$ | $\frac{9}{28}$  | $\frac{3}{28}$ | $\frac{15}{28}$ |
|        | 1 | $\frac{3}{14}$ | $\frac{3}{14}$  | 0              | $\frac{3}{7}$   |
|        | 2 | $\frac{1}{28}$ | 0               | 0              | $\frac{1}{28}$  |
| $g(x)$ |   | $\frac{5}{14}$ | $\frac{15}{28}$ | $\frac{3}{28}$ | 1               |

We need to calculate  $\sigma_X^2 = E[X^2] - \mu_X^2$  and  $\sigma_Y^2 = E[Y^2] - \mu_Y^2$ .

$$\sigma_X^2 = \frac{45}{112}, \quad \sigma_Y^2 = \frac{9}{28}$$

Therefore,

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \cdot \sigma_Y} = \frac{-\frac{9}{56}}{\sqrt{\left(\frac{45}{112}\right) \left(\frac{9}{28}\right)}} = -\frac{1}{\sqrt{5}}$$

The fraction  $X$  of male runners and the fraction  $Y$  of female runners who compete in marathon races are described by the joint density function

$$f(x, y) = \begin{cases} 8xy & 0 \leq y \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the correlation coefficient of  $X$  and  $Y$ .

We know that  $\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \cdot \sigma_Y}$ .

$$\sigma_{XY} = E[XY] - E[X]E[Y] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xyf(x, y) dx dy - \int_{-\infty}^{\infty} xg(x) dx \int_{-\infty}^{\infty} yh(y) dy$$

$$\sigma_X^2 = E[X^2] - (E[X])^2 = \int_{-\infty}^{\infty} x^2 g(x) dx - \left( \int_{-\infty}^{\infty} xg(x) dx \right)^2$$

$$\sigma_Y^2 = E[Y^2] - (E[Y])^2 = \int_{-\infty}^{\infty} y^2 h(y) dy - \left( \int_{-\infty}^{\infty} yh(y) dy \right)^2$$

$$f(x, y) = \begin{cases} 8xy & 0 \leq y \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$g(x) = \int_0^x f(x, y) dy, \quad h(y) = \int_y^1 f(x, y) dx$$

$$g(x) = \begin{cases} 4x^3 & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad h(y) = \begin{cases} 4y(1 - y^2) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 g(x) dx = \int_0^1 4x^5 dx = \frac{2}{3}, \text{ and } E[Y^2] = \frac{1}{3}.$$

$$E[X] = \int_{-\infty}^{\infty} x g(x) dx = \int_0^1 4x^4 dx = \frac{4}{5}, \text{ and } E[Y] = \frac{8}{15}.$$

$$\sigma_{XY} = E[XY] - E[X]E[Y] = \int_0^1 \int_y^1 8x^2 y^2 dx dy - \left(\frac{4}{5}\right) \left(\frac{8}{15}\right) = \frac{4}{225}$$

$$\sigma_X^2 = \frac{2}{3} - \left(\frac{4}{5}\right)^2 = \frac{2}{75} \text{ and } \sigma_Y^2 = \frac{11}{225}. \text{ Thus } \rho_{XY} = \frac{4}{66}.$$

- Marginal

$$g(x) = \begin{cases} \sum_y f(x, y) & \text{Dis.} \\ \int_{-\infty}^{\infty} f(x, y) dy & \text{Cont.} \end{cases} \quad h(y) = \begin{cases} \sum_x f(x, y) & \text{Dis.} \\ \int_{-\infty}^{\infty} f(x, y) dx & \text{Cont.} \end{cases}$$

- Conditional Probability  $f(y|x) = \frac{f(x,y)}{g(x)}$ , provided  $g(x) > 0$

- Expected value

$$E[g(X, Y)] = \begin{cases} \sum_x \sum_y g(x, y) f(x, y) & \text{Dis.} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) dx dy & \text{Cont.} \end{cases}$$

- Covariance  $\sigma_{XY} = E[XY] - \mu_X \mu_Y$ ,

$$\mu_X = \sum_x \sum_y x f(x, y) = \sum_x x g(x), \text{ and}$$

$$\mu_Y = \sum_x \sum_y y f(x, y) = \sum_y y h(y)$$

- Correlation  $\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$ ,  $\sigma_X^2 = E[X^2] - \mu_X^2$  and  $\sigma_Y^2 = E[Y^2] - \mu_Y^2$

From a sack of fruit containing 3 oranges, 2 apples, and 3 bananas, a random sample of 4 pieces of fruit is selected. If  $X$  is the number of oranges and  $Y$  is the number of apples in the sample, find

- (a) the joint probability distribution of  $X$  and  $Y$   
 (b)  $P[(X, Y) \in A]$ , where  $A$  is the region that is given by  $\{(x, y) | x + y \leq 2\}$

$$f(x, y) = \frac{C_x^3 C_y^2 C_{4-x-y}^3}{C_4^8}$$

|     | X=0            | X=1             | X=2             | X=3            |
|-----|----------------|-----------------|-----------------|----------------|
| Y=0 | 0              | $\frac{3}{70}$  | $\frac{9}{70}$  | $\frac{3}{70}$ |
| Y=1 | $\frac{2}{70}$ | $\frac{18}{70}$ | $\frac{18}{70}$ | $\frac{2}{70}$ |
| Y=2 | $\frac{3}{70}$ | $\frac{9}{70}$  | $\frac{3}{70}$  | 0              |

$$\begin{aligned}
 P(X + Y \leq 2) &= \sum_{y=0}^2 \sum_{x=0}^{2-y} f(x, y) \\
 &= (f(0, 0) + f(1, 0) + f(2, 0)) \\
 &\quad + (f(0, 1) + f(1, 1)) + f(0, 2) \\
 P(X + Y \leq 2) &=?
 \end{aligned}$$

The covariance of  $X$  and  $Y$  is  $\sigma_{XY} = E[XY] - \mu_X \mu_Y$

$$E[XY] = \sum_x \sum_y xyf(x, y) = \frac{9}{7}, \quad \mu_X = \sum_x \sum_y xf(x, y) = \frac{3}{2}, \quad \mu_Y = 1$$

Suppose that 3 balls are randomly selected from an urn containing 3 red, 4 white, and 5 blue balls. If we let  $X$  and  $Y$  denote, respectively, the number of red and white balls chosen, then the joint probability mass function of  $X$  and  $Y$

$$f(x, y) = \frac{C_x^3 C_y^4 C_{3-x-y}^5}{C_3^{12}}$$

|     | X=0    | 1      | 2      | 3     |
|-----|--------|--------|--------|-------|
| Y=0 | 10/220 | 30/220 | 15/220 | 1/220 |
| 1   | 40/220 | 60/220 | 12/220 | 0     |
| 2   | 30/220 | 18/200 | 0      | 0     |
| 3   | 4/220  | 0      | 0      | 0     |

| x                                                                                         | 0      | 1       | 2      | 3     |
|-------------------------------------------------------------------------------------------|--------|---------|--------|-------|
| g(x)                                                                                      | 84/220 | 108/220 | 27/220 | 1/220 |
| $\mu_X = \sum_x xg(x) = \frac{108}{220} + 2 \cdot \frac{27}{220} + 3 \cdot \frac{1}{220}$ |        |         |        |       |

| y                                                                                         | 0      | 1       | 2      | 3     |
|-------------------------------------------------------------------------------------------|--------|---------|--------|-------|
| h(y)                                                                                      | 56/220 | 112/220 | 48/220 | 4/220 |
| $\mu_Y = \sum_y yh(y) = \frac{112}{220} + 2 \cdot \frac{48}{220} + 3 \cdot \frac{4}{220}$ |        |         |        |       |



The covariance between two random variables is a measure of the nature of the association between the two.

- If large values of  $X$  often result in large values of  $Y$  or small values of  $X$  result in small values of  $Y$
- positive  $X - \mu_X$  will often result in positive  $Y - \mu_Y$  and negative  $X - \mu_X$  will often result in negative  $Y - \mu_Y$

Thus, the product  $(X - \mu_X)(Y - \mu_Y)$  will tend to be positive.

On the other hand, if large  $X$  values often result in small  $Y$  values, the product  $(X - \mu_X)(Y - \mu_Y)$  will tend to be negative.

Covariance measures how the two variables move concerning each other and is an extension of the concept of variance (which tells about how a single variable varies). It can take any value from  $-\infty$  to  $\infty$ .

- The higher this value, the more dependent the relationship is. A positive number signifies positive covariance and denotes a direct connection. Effectively this means that an increase in one variable would also lead to a corresponding increase in the other variable, provided other conditions remain constant.
- A negative number signifies negative covariance, which denotes an inverse relationship between the two variables.

Though covariance is perfect for defining the type of relationship, it is not good for interpreting its magnitude.

Correlation is a step ahead of covariance as it quantifies the relationship between two random variables. In simple terms, it is a unit measure of how these variables change concerning each other (normalized covariance value).

- The correlation has an upper and lower cap on a range, unlike covariance. It can only take values between  $+1$  and  $-1$ . A correlation of  $+1$  indicates that random variables have a direct and strong relationship.
- On the other hand, the correlation of  $-1$  indicates a strong inverse relationship, and an increase in one variable will lead to an equal and opposite decrease in the other variable.  $0$  means that the two numbers are independent.